

Crypto challenge - Mersenne seed recovery

deuterium — 2021-07-25

Canonical HTML: <https://deut-erium.github.io/2021/07/25/mersenne-seed-recovery.html>

Abstract

Crypto challenge - Mersenne seed recovery, published in deuterium's blog.

Can array based initialization of mersenne twister which is used in current libraries be reversed to get the seed (any 1-19937 bit quantity)

I hope its the time to dive into the mersenne twister implementation of python to get a hang of `init_by_array` seed initialization. I give you the first 32 bit outputs from the random number, you go tell me what the seed was i.e your flag.

```
from secret import flag
import random

seed = int.from_bytes(flag, 'big')
random.seed(seed)

outputs = [random.getrandbits(32) for i in range(624)]
print(outputs)

# outputs
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Enter the flag

The local SHA-256 checker requires JavaScript. The challenge itself does not.

Interactive controls are omitted from this PDF. See the canonical HTML version.